Controlled Gradient-Flow Ablation in Deep Multilayer Perceptrons

A Reproducible Case Study of Activation, Initialization, and Depth

Rajesh Phulwaria
Independent Researcher
India
phulwariarajesh152@gmail.com

November 2026

GitHub: RajeshPhulwaria006 **LinkedIn:** Rajesh Phulwaria

Abstract

Gradient propagation is an important factor in the optimization of deep neural networks. When gradients become extremely small, early layers may receive weak learning signals, making optimization difficult. This case study investigates gradient behavior in multilayer perceptrons under controlled experimental conditions.

Three ablation studies were conducted using PyTorch: (1) activation-function ablation comparing Sigmoid, Tanh, and ReLU; (2) depth scaling using ReLU with He initialization; and (3) initialization ablation comparing Xavier and He initialization for ReLU networks.

Each configuration was evaluated using three random seeds. A single synthetic mini-batch with random binary targets was used to generate a reproducible backward-pass diagnostic. The L2 norm and RMS of each linear layer's weight gradient were recorded immediately after backpropagation.

The results show clear differences in gradient magnitude between activation functions and initialization schemes. In particular, the earliest hidden layer under Sigmoid activation had a mean gradient L2 norm of approximately 2.16×10^{-6}, compared with 1.09×10^{-1} for Tanh and 1.74×10^{-2} for ReLU under the same Xavier-initialized setting. For the ReLU initialization comparison, the earliest-layer mean gradient was approximately 2.28×10^{-1} with He initialization and 1.74×10^{-2} with Xavier initialization.

Increasing depth also changed the gradient profile. The earliest-layer mean gradient decreased from approximately 2.54×10^{-1} at depth 4 to 1.12×10^{-1} at depth 32 under the tested ReLU/He configuration.

These results demonstrate that activation function, initialization, and depth can substantially influence gradient magnitudes. However, because the experiment uses a single synthetic backward-pass diagnostic rather than a real learning task, the results should not be interpreted as evidence that one configuration universally trains better than another.

Keywords: gradient flow, vanishing gradients, neural networks, ablation study, ReLU, Sigmoid, Tanh, Xavier initialization, He initialization, deep learning

1 Introduction

Training a deep neural network depends on the ability of information and gradients to propagate through multiple layers. During backpropagation, the gradient received by an early layer depends on a sequence of derivatives and weight transformations. If these transformations repeatedly reduce the gradient magnitude, the resulting gradient may become very small.

This phenomenon is commonly associated with the vanishing-gradient problem. It is particularly important for deep feedforward networks because a weak gradient in an early layer can reduce the effectiveness of parameter updates.

The purpose of this study is not to demonstrate a new neural-network architecture. Instead, the study focuses on a simpler research question:

> *Under controlled experimental conditions, how do activation function, weight initialization, and network depth affect the magnitude and distribution of layer-wise gradients in a multilayer perceptron?*

The main research principle is controlled comparison. Rather than changing several components simultaneously, each study varies one primary factor while keeping the remaining experimental conditions fixed.

This approach is useful because a difference in gradient behavior can then be more directly associated with the factor being investigated.

2 Motivation and Relation to Previous Work

This study continues an experimental investigation of gradient behavior in deep neural networks. The author's earlier work examined the vanishing-gradient problem using different neural-network configurations.

The present study narrows the experimental question further by using controlled ablations. In particular, activation function, initialization, and depth are investigated separately rather than treating a complete neural-network configuration as a single combined treatment.

The objective is therefore methodological as much as experimental:

- isolate individual experimental factors;
- measure layer-wise gradient behavior;
- quantify variation across multiple random seeds;
- make the experiment reproducible;
- distinguish direct observations from broader interpretations.

This work should be viewed as a reproducible experimental case study rather than a claim of a novel optimization method.

3 Research Questions

The study addresses three main questions.

3.1 RQ1: Activation

How does the choice of activation function affect layer-wise weight-gradient magnitude when network depth and Xavier initialization are fixed?

3.2 RQ2: Depth

How does increasing the number of hidden layers affect the distribution of weight-gradient magnitudes when ReLU activation and He initialization are fixed?

3.3 RQ3: Initialization

How does Xavier initialization compare with He initialization for a ReLU network when depth and activation are fixed?

4 Experimental Design

The model is a fully connected multilayer perceptron (MLP). The hidden layers use a configurable activation function, while the final linear layer produces a single logit.

The baseline architecture uses:

- input dimension: 20;
- hidden width: 64;
- variable hidden depth;
- one scalar output logit;
- binary cross-entropy with logits loss;
- batch size: 128;
- random binary targets;
- random seeds: 11, 23, and 37.

The gradient measurement is performed immediately after:

$$\text{forward pass} \rightarrow \text{loss computation} \rightarrow \text{backward pass}.$$

No optimizer update is performed.

This makes the experiment a diagnostic of gradient propagation rather than a full training benchmark.

5 Model Architecture

For an input vector $\mathbf{x}$, each hidden layer performs

$$\mathbf{h}^{(l)} = \phi\left(W^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right),$$

where ϕ is the selected activation function.

The final layer produces a scalar logit:

$$z = W^{(L)}\mathbf{h}^{(L-1)} + b^{(L)}.$$

The binary cross-entropy loss with logits is then computed as

$$\mathcal{L} = \text{BCEWithLogitsLoss}(z, y).$$

After computing the loss, PyTorch automatic differentiation is used to obtain the gradient of every linear layer.

6 Gradient Metrics

For every linear layer, two gradient measurements were recorded.

6.1 L2 Gradient Norm

For a parameter-gradient vector g,

$$\|g\|_2 = \sqrt{\sum_i g_i^2}.$$

A smaller value indicates a smaller aggregate gradient for that layer and batch.

The L2 norm is useful for observing how gradient magnitude changes across layers.

6.2 Gradient RMS

The root-mean-square gradient is calculated as

$$\mathrm{RMS}(g) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} g_i^2}.$$

RMS is useful because the L2 norm can depend on the number of parameters in a layer. Therefore, both metrics are recorded.

7 Experimental Conditions

Table 1: Controlled experimental studies.

Study	Main Factor	Fixed Conditions	Settings
A	Activation	Depth=8, Xavier	Sigmoid/Tanh/ReLU
B	Depth	ReLU, He	4/8/16/32 layers
C	Initialization	ReLU, Depth=8	Xavier/He

Each configuration was evaluated using the three seeds

$$\{11, 23, 37\}.$$

This gives a small estimate of variation caused by random initialization and random synthetic data generation.

8 Study A: Activation-Function Ablation

8.1 Objective

Study A investigates how activation functions affect gradient propagation while keeping network depth and initialization fixed.

The network contains eight hidden layers and uses Xavier initialization. The three activation functions are:

- Sigmoid;
- Tanh;
- ReLU.

8.2 Results

The earliest hidden layer provides a useful comparison because it is furthest from the loss during backpropagation.

Table 2: Study A: mean gradient L2 norm at the earliest hidden layer.

Activation	Mean Gradient L2
Sigmoid	2.16×10^{-6}
Tanh	1.09×10^{-1}
ReLU	1.74×10^{-2}

The Sigmoid configuration produced an extremely small gradient at the earliest hidden layer. Its mean gradient L2 norm was approximately

$$2.16 \times 10^{-6}.$$

In comparison, Tanh produced approximately

$$1.09 \times 10^{-1},$$

while ReLU produced approximately

$$1.74 \times 10^{-2}.$$

Therefore, under this specific experimental configuration, the earliest-layer gradient for Tanh was approximately 5.0×10^4 times larger than the corresponding Sigmoid value.

The important observation is not the ratio itself, but the large difference in gradient scale between the activation functions.

8.3 Layer-Wise Behavior

The Sigmoid configuration shows a strong increase in gradient magnitude as the measurement moves toward the output layer. For example, its mean gradient L2 increases from approximately

$$2.16 \times 10^{-6}$$

at layer 0 to

$$6.13 \times 10^{-2}$$

at the final hidden linear layer (layer 7).

The output layer has a larger gradient of approximately

$$1.93 \times 10^{-1}.$$

The Tanh profile remains comparatively stable across the hidden layers, with values roughly between 1.0×10^{-1} and 1.5×10^{-1}.

The ReLU/Xavier profile remains in the approximate range of 1.7×10^{-2} to 2.4×10^{-2} across the hidden layers.

These observations suggest that activation choice strongly affects the scale and distribution of parameter gradients in the tested network.

8.4 Activation Ablation Figure

The recommended figure is a logarithmic plot of mean layer-wise gradient L2 with standard deviation across seeds.

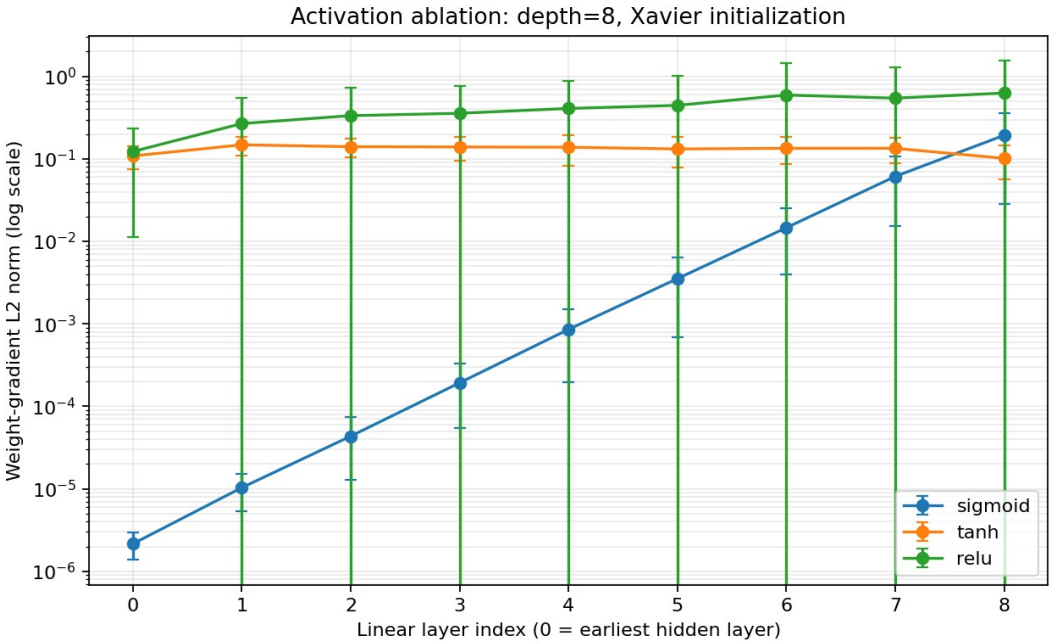


Figure 1: Activation ablation for an eight-hidden-layer MLP with Xavier initialization. The y-axis uses a logarithmic scale. Error bars represent standard deviation across the three seeds.

Figure note: The activation plot should be generated only from the Study A condition: depth=8 and initialization=Xavier. The original plotting implementation filtered only by depth and therefore mixed ReLU/He observations from the other studies. The corrected figure should be used in the final manuscript.

9 Study B: Depth Scaling

9.1 Objective

Study B investigates how increasing network depth changes gradient behavior.

The activation function is fixed to ReLU and the initialization method is fixed to He initialization. Four depths are evaluated:

$$4,\ 8,\ 16,\ 32.$$

9.2 Results

The mean gradient L2 norm at the earliest hidden layer is shown below.

Table 3: Study B: earliest-layer gradient magnitude across network depths.

Depth	Mean Gradient L2 at Layer 0
4	0.2542
8	0.2284
16	0.2155
32	0.1123

The earliest-layer gradient decreases as depth increases in this particular experiment. The measured values are approximately:

$$0.2542, \quad 0.2284, \quad 0.2155, \quad 0.1123$$

for depths 4, 8, 16, and 32 respectively.

The depth-32 network therefore has the smallest earliest-layer gradient among the four tested configurations.

However, the complete layer-wise profiles show that gradient behavior is not simply a monotonic decay across every layer.

For example, the depth-16 model reaches mean gradient L2 values above 2.0 in several later layers, while the depth-32 model remains much closer to the 0.2–0.5 range over most of its layers.

This shows why a single gradient value is insufficient to characterize gradient flow in a deep network.

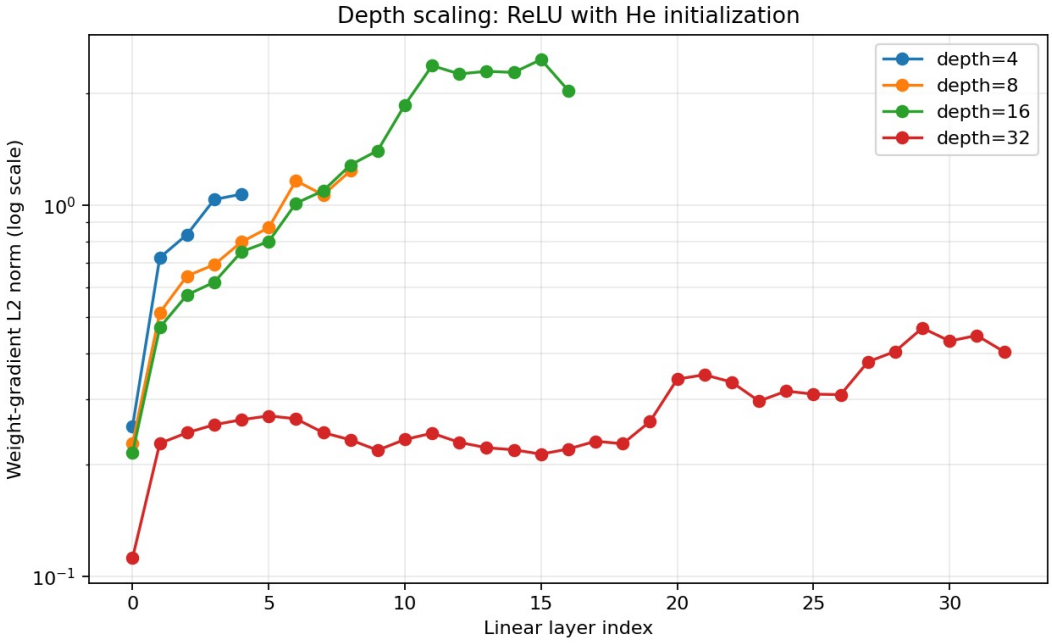


Figure 2: Depth scaling under ReLU activation and He initialization. Each curve represents the mean weight-gradient L2 norm across three seeds. The y-axis is logarithmic.

The depth figure demonstrates that changing depth alters the complete spatial distribution of gradient magnitude across the network.

Because the networks have different numbers of layers, comparisons based only on the same forward layer index should be interpreted carefully. A stronger follow-up analysis would align layers by distance from the output or by relative network depth.

10 Study C: Initialization Ablation

10.1 Objective

Study C examines the effect of initialization while keeping the activation function and network depth fixed.

The network uses:

- ReLU activation;
- eight hidden layers;
- Xavier or He initialization.

10.2 Results

At the earliest hidden layer, the mean gradient L2 values are:

Table 4: Study C: initialization comparison at the earliest hidden layer.

Initialization	Mean Gradient L2
Xavier	0.0174
He	0.2284

He initialization produces a substantially larger gradient magnitude than Xavier initialization in the tested ReLU network.

The earliest-layer values are approximately

$$0.2284$$

for He and

$$0.0174$$

for Xavier.

The He value is approximately 13.1 times larger than the Xavier value.

The difference continues through most of the hidden layers. For example, at layer 6 the mean gradient L2 is approximately

$$1.1651$$

with He initialization compared with

$$0.0237$$

with Xavier initialization.

This indicates that initialization has a strong effect on gradient scale under the tested conditions.

11 Summary Across Seeds

For each experimental condition and layer, the raw measurements were aggregated across the three seeds.

The summary contains:

- mean gradient L2;
- standard deviation of gradient L2;
- mean gradient RMS;
- mean loss.

The standard deviation provides an estimate of how sensitive the measured gradient magnitude is to the random seed.

For example, in Study A, the earliest-layer Sigmoid gradient has a mean of approximately 2.16×10^{-6} and a standard deviation of approximately 1.0×10^{-6}.

The corresponding ReLU/Xavier value is approximately 1.74×10^{-2} with a standard deviation of approximately 6.25×10^{-4}.

This relatively small variation for the ReLU/Xavier earliest layer indicates that the observed difference is not caused by a single unusual seed.

Nevertheless, three seeds remain a small sample and should not be treated as a strong statistical estimate.

12 Loss Behavior

The loss values are included as a diagnostic measurement.

The mean losses for the main configurations were approximately:

Table 5: Mean loss for the tested configurations.

Depth	Activation	Initialization	Mean Loss
8	Sigmoid	Xavier	0.6955
8	Tanh	Xavier	0.7043
8	ReLU	Xavier	0.6940
4	ReLU	He	0.8766
8	ReLU	He	0.8149
16	ReLU	He	0.8901
32	ReLU	He	0.7276

These losses should not be interpreted as training performance because the experiment performs only a single forward/backward pass and uses random binary targets.

The loss is therefore useful as a record of the diagnostic state rather than as evidence that one architecture learns better.

13 Discussion

The three studies provide complementary observations.

13.1 Activation Function

The activation comparison produces the clearest evidence of different gradient scales.

Sigmoid produces extremely small gradients in the earliest hidden layer. Tanh produces substantially larger gradients, while ReLU also maintains gradients at a much larger scale than Sigmoid in this configuration.

This behavior is consistent with the mathematical role of activation derivatives during backpropagation.

However, the experiment does not establish that Tanh is universally better than ReLU or that Sigmoid cannot be trained successfully. The observation is specific to the tested architecture, initialization, data, loss, and single backward pass.

13.2 Depth

Increasing depth changes the gradient profile.

The earliest-layer gradient decreases from approximately 0.254 at depth 4 to approximately 0.112 at depth 32. At the same time, the complete gradient profile can contain large variations across later layers.

Therefore, depth should not be described simply as "causing vanishing gradients." The result is more accurately stated as:

> Increasing depth changed the distribution of gradient magnitudes under the tested ReLU/He configuration, including a reduction in the earliest-layer gradient at depth 32.

13.3 Initialization

The initialization comparison shows a substantial difference between Xavier and He initialization for the ReLU network.

He initialization produces larger gradient magnitudes throughout much of the network. This is consistent with the motivation behind variance-preserving initialization schemes for rectifier-based networks.

Again, the experiment measures gradient magnitude rather than final learning quality. A larger gradient is not automatically a better gradient.

14 Main Findings

The main empirical findings are:

1. **Activation matters.** Under depth 8 and Xavier initialization, Sigmoid produced an earliest-layer mean gradient L2 of approximately 2.16×10^{-6}, substantially below Tanh and ReLU.
2. **Depth changes gradient distribution.** Under ReLU/He, the earliest-layer gradient decreased from approximately 0.254 at depth 4 to 0.112 at depth 32.
3. **Initialization strongly affects gradient scale.** At depth 8 with ReLU, the earliest-layer gradient was approximately 0.228 with He initialization compared with 0.0174 with Xavier initialization.
4. **Gradient behavior is layer-dependent.** Later layers can exhibit substantially different gradient magnitudes from earlier layers. Therefore, a single network-level gradient statistic is insufficient to describe gradient propagation.
5. **Seed variation exists.** The standard deviations show that gradient measurements vary across random seeds. More repetitions would be required for stronger statistical claims.

15 Reproducibility

The experiment records the following metadata for every measurement:

- random seed;

- network depth;
- activation function;
- initialization method;
- linear-layer index;
- output-layer indicator;
- weight-gradient L2 norm;
- weight-gradient RMS;
- loss.

The complete experiment produced 327 layer-level measurements across the three studies.

The aggregated summary contains 91 condition-layer rows.

The experiment was designed to run on CPU and can also be executed in a Google Colab environment.

The controlled setup and recorded seeds make it possible to repeat the experiment and compare future modifications against the current baseline.

16 Limitations

Several limitations should be considered before drawing broader conclusions.

16.1 Synthetic Data

The experiment uses randomly generated inputs and random binary targets. This creates a reproducible backward-pass signal but does not represent a meaningful supervised learning task.

Therefore, the results describe gradient behavior under the experimental setup rather than performance on a real dataset.

16.2 Single Backward Pass

The study measures gradients after one backward pass.

It does not measure:

- convergence;
- training accuracy;
- validation accuracy;
- generalization;
- long-term optimization stability.

16.3 Limited Number of Seeds

Only three seeds were used.

This is sufficient for a small reproducible case study, but a larger number of seeds would provide a stronger estimate of variability.

16.4 Output-Layer Initialization

The initialization routine applies the selected initialization scheme to all linear layers, including the output layer.

This is acceptable for the current diagnostic experiment but is not necessarily the best task-specific output-layer initialization.

A refined study should separate hidden-layer initialization from output-layer initialization.

16.5 Parameter Gradients Only

The current study measures parameter-gradient norms.

Activation-gradient measurements could provide an additional view of how the backpropagated signal changes through hidden representations.

16.6 Depth Comparison

The depth experiment plots networks with different numbers of layers using forward layer index.

This makes direct comparison between the same numerical layer index imperfect. A future version should also align layers relative to the output layer.

17 Data and Figure Integrity

An important implementation detail was identified during analysis of the generated figures.

The original activation plotting function selected all rows satisfying

$$\texttt{depth == 8}.$$

However, the experiment contains both ReLU/Xavier and ReLU/He configurations at depth 8. Therefore, the original plot mixes observations from Study A with observations from Studies B and C.

As a result, the supplied activation figure should not be used as the final Study A figure.

The numerical results reported in this paper are based on the grouped Study A condition:

$$\text{depth} = 8, \qquad \text{initialization} = \text{Xavier}.$$

The corrected plotting condition should therefore explicitly select:

$$\texttt{depth == 8} \quad \wedge \quad \texttt{init == "xavier"}.$$

This correction is important for maintaining a valid controlled comparison.

The issue does not invalidate the underlying raw measurements or the depth-scaling experiment, but it does affect the original activation visualization.

18 Future Work

Several extensions can make this study more rigorous.

18.1 Real Dataset

The first extension should replace random targets with a small real classification dataset.

This would allow gradient-flow observations to be connected with actual optimization and generalization behavior.

18.2 Training Dynamics

Instead of measuring only one backward pass, gradients could be recorded over multiple training epochs.

This would allow analysis of:

- gradient magnitude over time;
- loss convergence;
- training stability;
- validation performance.

18.3 Activation-Gradient Measurement

The next version should also record the gradient of hidden activations:

$$\left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(l)}} \right\|_2 .$$

This would complement parameter-gradient measurements.

18.4 Factorial Design

A larger controlled experiment could investigate interactions between:

- activation;
- initialization;
- depth;
- width.

This would allow the study to move beyond one-factor-at-a-time comparisons.

19 Conclusion

This case study investigated gradient flow in multilayer perceptrons under controlled variations of activation function, network depth, and weight initialization.

The experiments show that activation function and initialization can produce large differences in layer-wise gradient magnitude. In particular, Sigmoid produced extremely small gradients in the earliest hidden layer under the tested Xavier configuration, while Tanh and ReLU produced substantially larger values.

Increasing depth also changed gradient behavior. Under ReLU with He initialization, the earliest-layer mean gradient decreased from approximately 0.254 at depth 4 to 0.112 at depth 32. However, the complete profiles showed that gradient behavior across a deep network is not a simple monotonic process.

The initialization study further showed that He initialization produced much larger gradient magnitudes than Xavier initialization for the tested ReLU network.

The central conclusion is therefore:

> **Gradient propagation is strongly dependent on the interaction between network depth, activation function, and initialization strategy. Controlled experiments are necessary to separate these effects and avoid attributing observed behavior to the wrong factor.**

At the same time, the results should be interpreted as a gradient-flow diagnostic rather than as a complete evaluation of neural-network training. The use of synthetic data, random targets, a single backward pass, and three seeds limits the scope of the conclusions.

The value of this study is therefore its controlled experimental methodology, reproducible measurements, and explicit separation between observation and interpretation.

Research Statement

This work represents an independent experimental investigation into the behavior of neural-network gradients. The study was designed, implemented, executed, analyzed, and documented as an individual research project.

The emphasis is on reproducible experimentation: defining a controlled question, implementing the experiment, recording layer-level measurements, inspecting variation across random seeds, identifying implementation issues, and separating measured observations from theoretical expectations.

The study is intentionally presented as a case study rather than as a claim of a new optimization method or universal empirical law.

References

[1] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2010.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.

[3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

[4] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems*, 2019.

A Experimental Configuration

Table 6: Complete baseline configuration.

Parameter	Value
Input dimension	20
Hidden width	64
Batch size	128
Loss	BCEWithLogitsLoss
Activations	Sigmoid, Tanh, ReLU
Initializations	Xavier, He
Depths	4, 8, 16, 32
Gradient metric	Weight-gradient L2 and RMS
Dataset	Synthetic random inputs/targets